034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

Second Partial Examination - A.Y. 2024/2025

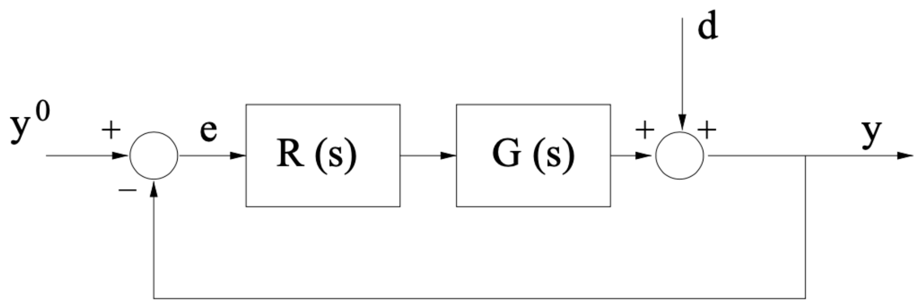
June 6, 2025

Design of a Strictly Proper Controller for a Plant with a Parameter Affected by Interval Uncertainty

Table of Contents

Description of the Feedback Control System

Consider the feedback control system described by the block-scheme shown in the following figure



where $G(s)$ is given by

$$G(s) = 9\,\frac{\left(1+\frac{s}{10}\right)}{(1+\tau\, s)\left(1+\frac{s}{256}\right)}\,, \qquad \tau \in \left[\frac{1}{2},\, 10\right]\ \text{sec}$$

Note: This means that the system has a striclty proper transfer function with two distinct, real poles, and only one zero. One of the poles is exactly known, whereas the second one is partially unknown. The time constant τ of this pole takes on a value contained within the assigned range $[0.5\,,\ 10]$ sec.

Question

Q1: Design a **strictly proper controller** $C(s)$ such that all following requirements are simultaneously met:

- Zero steady-state error in unit step response (closed-loop) $e(\infty) = 0$, against disturbances given by

$$\begin{cases} y^o(t) &= 1(t) \\ d(t) &= A \cdot 1(t)\,,\ A \in \mathbb{R} \end{cases} \implies e_\infty = 0$$

where $\mathbf{1}(t)$ denotes the unit step function;

- phase margin greater than or equal to 40°: $\varphi_m \geq 40^\circ$;
- settling time [at 1% as usual] in the (closed-loop) unit step output less than or equal to 10 seconds: $t_{s\,1\%} \leq 10$ sec

Hint

How to handle the unknown time constant τ? A good practice is to identify the **worst-case scenario** and design the controller $C(s)$ according to that scenario.

Solution

Initialisation code (clearing the MATLAB workspace, adding the path of the BodeDiagram subfolder to the MATLAB search-path)

```
clear variables

% adding folders (and subfolders) to search path
addpath(genpath('BodeDiagram/'))

% let's define the transfer function builder element
s=tf('s');
```

Preliminary Analysis

$$G(s) = 9\,\frac{\left(1+\frac{s}{10}\right)}{(1+\tau\,s)\left(1+\frac{s}{256}\right)}\,, \qquad \tau \in \left[\frac{1}{2},\,10\right]\ \text{sec}$$

According to (Q1.1), the regulator must have a pole at $s=0$: $\quad C_1(s) = \frac{\mu_C}{s}\,, \quad \mu_C > 0$

```
tauMIN = 0.5; tauMAX = 10;
tauR = 4.5;

% static regulator, according to (Q1,1)
muC = 1;
C1s = muC/s;

% what is the worst-case scenario?
Gs1 = 9*(1+s/10)/(1+tauMIN*s)/(1+s/256);
Gs2 = 9*(1+s/10)/(1+tauMAX*s)/(1+s/256);
Gs3 = 9*(1+s/10)/(1+tauR*s)/(1+s/256);

Ls1 = C1s*Gs1;
Ls2 = C1s*Gs2;
Ls3 = C1s*Gs3;
```

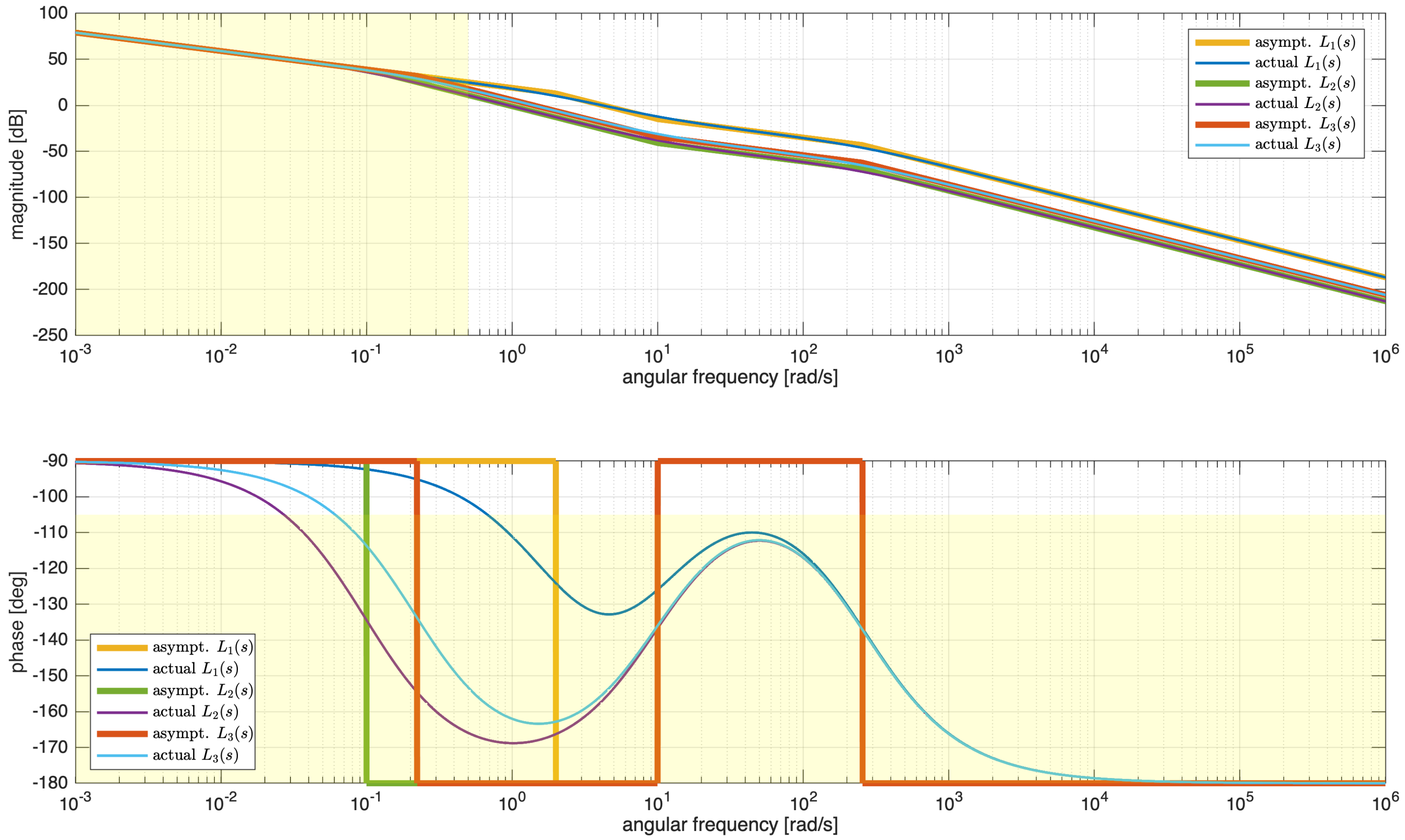
Let's analyse the (Q1.3) requirement:

$$t_{s\,1\%} \leq 10 \quad \Longleftrightarrow \quad \begin{cases} \varphi_m > 75^\circ \implies t_{s\,1\%} = \dfrac{4.6}{\omega_c} \leq 10 & \implies \quad \omega_c \geq \dfrac{46}{100} \\ \varphi_m < 75^\circ \implies t_{s\,1\%} = \dfrac{4.6}{\xi\,\omega_n} \approx \dfrac{4.6}{\frac{\varphi_m}{100}\omega_c} \leq 10 & \implies \varphi_m \cdot \omega_c \geq 46 \end{cases} \qquad (*)$$

Consider the first scenario ($\varphi_m > 75^\circ$):

```
% the Bode diagrams

Bcolors = [0 0.4470 0.7410; ...
           0.9290 0.6940 0.1250; ...
           0.4940 0.1840 0.5560; ...
           0.4660 0.6740 0.1880; ...
           0.3010 0.7450 0.9330; ...
           0.8500 0.3250 0.0980; ...
           0.6350 0.0780 0.1840]; % some different colors


hf = figure('Units','normalized','Position',[0.1, 0.1, 0.95, 0.85]);
drawBodediagrams(Ls1, [], Bcolors(2,:) , 3.5, [], Bcolors(1,:) ,1.5, [], hf);
drawBodediagrams(Ls2, [], Bcolors(4,:) , 3.5, [], Bcolors(3,:) ,1.5, [], hf);
[ha1, ha2] = drawBodediagrams(Ls3, [], Bcolors(6,:) , 3.5, [], Bcolors(5,:) ,1.5, [], hf);

phaseM = +75; % the desired minimum phase margin, according to (Q1.2)&(Q1.3)
feas_Ls_Phase = -180+phaseM; % the minimum value of the arg L(j omega)
                             % to guarantee the feasibility of the design

YLIMS = ylim(ha2);
OMLIMs = xlim(ha2);
% the (Q1.2) constraint
```

```
minOmega1 = OMLIMs(1);
maxOmega1 = OMLIMs(2);
PhLIM = feas_Ls_Phase;

Vert1 = [minOmega1 ,  YLIMS(1); minOmega1 ,PhLIM; ...
         maxOmega1 , PhLIM; maxOmega1 , YLIMS(1)];
Faces1 = [1  2  3  4];
hold on;
patch(ha2, 'Faces',Faces1,'Vertices',Vert1,...
        'FaceColor','yellow','FaceAlpha',.15, ...
        'EdgeColor', 'none'); % the pale yellow region is the forbidden one

omC1 = 0.5; % constraint (Q1.3)&(Q1.2)
YLIMS = ylim(ha1);
OMLIMs = xlim(ha1);
minOmega1 = OMLIMs(1);
maxOmega1 = OMLIMs(2);
Vert2 = [minOmega1 ,  YLIMS(1); omC1 ,YLIMS(1); ...
         omC1 , YLIMS(2); minOmega1 , YLIMS(2)];
Faces2 = [1  2  3  4];
hold on;
patch(ha1, 'Faces',Faces2,'Vertices',Vert2,...
        'FaceColor','yellow','FaceAlpha',.15, ...
        'EdgeColor', 'none'); % the pale yellow region is the forbidden one

legend(ha1, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
              'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
              'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
              'Interpreter', 'latex', 'Location', 'best')
legend(ha2, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
              'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
              'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
              'Interpreter', 'latex', 'Location', 'southwest')
```



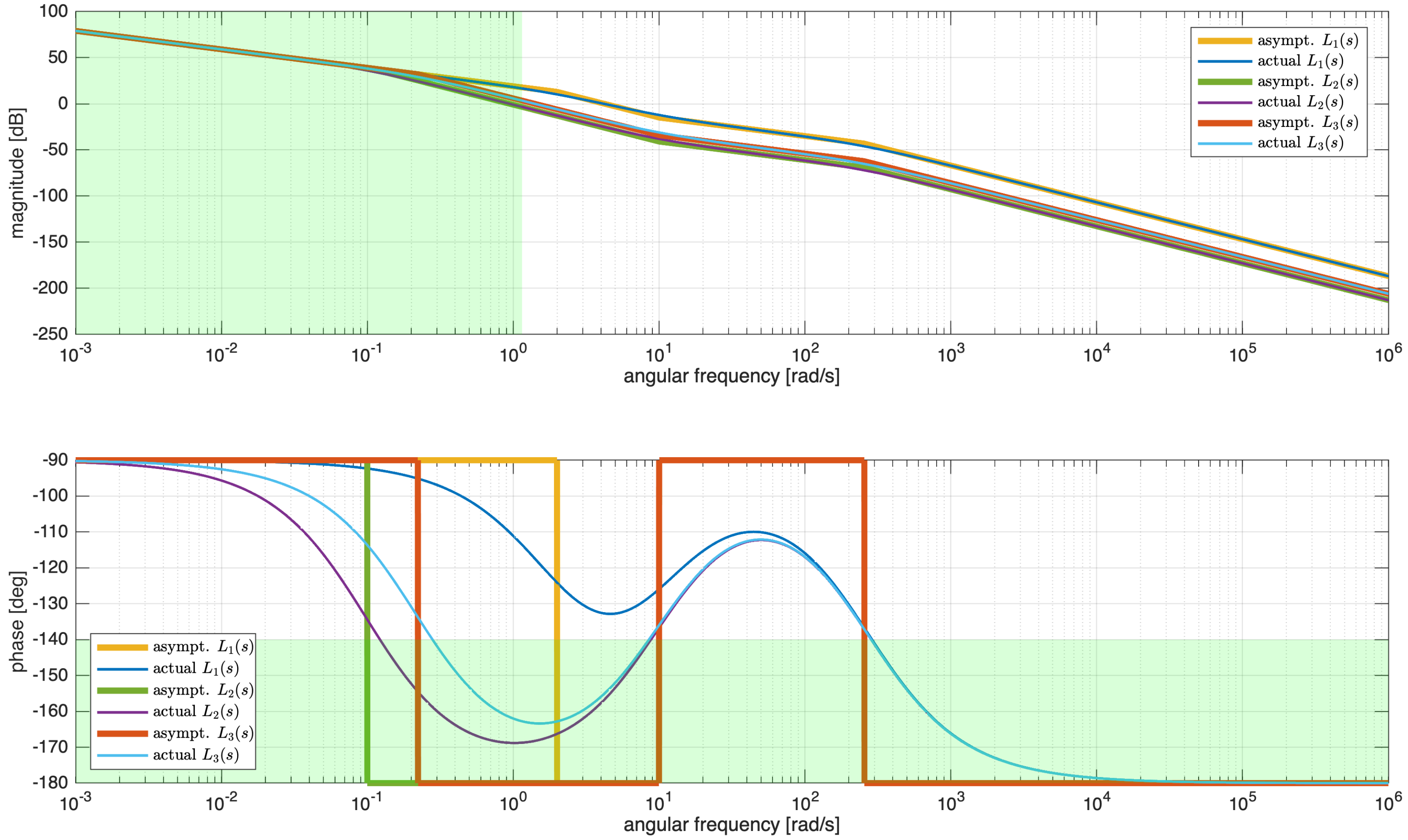
Without any modification to the controller $C(s)$ [i.e., without adding at least one zero in the second contribution to the controller $C_2(s)$], there is no way to fulfil the requirements (Q1.2) and (Q1.3).

What about the second scenario in Eq. $(*)$? Consider the minimum allowable value for the phase margin:

$$\varphi_m = 40^\circ \quad \Longrightarrow \quad \omega_c \geq \frac{46}{40} = 1.15 \text{ rad/sec}$$

```
hf = figure('Units','normalized','Position',[0.1, 0.1, 0.95, 0.85]);
drawBodediagrams(Ls1, [], Bcolors(2,:) , 3.5, [], Bcolors(1,:) ,1.5, [], hf);
drawBodediagrams(Ls2, [], Bcolors(4,:) , 3.5, [], Bcolors(3,:) ,1.5, [], hf);
[ha1, ha2] = drawBodediagrams(Ls3, [], Bcolors(6,:) , 3.5, [], Bcolors(5,:) ,1.5, [], hf);

phaseM = +40; % the desired minimum phase margin, according to (Q1.2)&(Q1.3)
feas_Ls_Phase = -180+phaseM; % the minimum value of the arg L(j omega)
                             % to guarantee the feasibility of the design

YLIMS = ylim(ha2);
OMLIMs = xlim(ha2);
% the (Q1.2) constraint
minOmega1 = OMLIMs(1);
```

```
maxOmega1 = OMLIMs(2);
PhLIM = feas_Ls_Phase;

Vert1 = [minOmega1 ,  YLIMS(1); minOmega1 ,PhLIM; ...
         maxOmega1 , PhLIM; maxOmega1 , YLIMS(1)];
Faces1 = [1  2  3  4];
hold on;
patch(ha2, 'Faces',Faces1,'Vertices',Vert1,...
        'FaceColor','green','FaceAlpha',.15, ...
        'EdgeColor', 'none'); % the pale yellow region is the forbidden one

omC1 = 1.15; % constraint (Q1.3)&(Q1.2)
YLIMS = ylim(ha1);
OMLIMs = xlim(ha1);
minOmega1 = OMLIMs(1);
maxOmega1 = OMLIMs(2);
Vert2 = [minOmega1 ,  YLIMS(1); omC1 ,YLIMS(1); ...
         omC1 , YLIMS(2); minOmega1 , YLIMS(2)];
Faces2 = [1  2  3  4];
hold on;
patch(ha1, 'Faces',Faces2,'Vertices',Vert2,...
        'FaceColor','green','FaceAlpha',.15, ...
        'EdgeColor', 'none'); % the pale yellow region is the forbidden one

legend(ha1, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
              'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
              'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
              'Interpreter', 'latex', 'Location', 'best')
legend(ha2, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
              'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
              'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
              'Interpreter', 'latex', 'Location', 'southwest')
```



This time, the design problem is feasible, with the simple controller $C_1(s) = \frac{\mu_C}{s}$ if we plan to obtain the crossover angular frequency in the interval $\omega_c \in [20\,,\,100]$ rad/sec.

The worst-case scenario corresponds to the highest value for the time constant τ, i.e., $\tau = 10$.

Tuning the Controller

According to the previously performed analysis, any angular frequency $\omega \in [20\,,\,100]$ rad/sec allows to fulfil all the requirements.

Let's tune the gain constant μ_C with the aim to obtain $\bar{\omega} = 20$ rad/sec as crossover frequency in the worst-case scenario

$$\begin{cases} C_1(s) &= \dfrac{\mu_C}{s} \\ G_2(s) &= 9\,\dfrac{\left(1+\frac{s}{10}\right)}{(1+\tau\,s)\left(1+\frac{s}{256}\right)}\,, \qquad \tau = 10 \text{ sec} \end{cases} \implies L_2(s) = 9\,\mu_C\,\frac{\left(1+\frac{s}{10}\right)}{s\,(1+10\,s)\left(1+\frac{s}{256}\right)}$$

Thus

$$\omega_c = 20 \quad \Longrightarrow \quad |L_2(j20)| = 1 = 9\,\mu_C \left| \frac{\left(1+\frac{j\,20}{10}\right)}{j20\,(1+10\cdot j20)\left(1+\frac{j20}{256}\right)} \right| \quad \Longrightarrow \quad \mu_C = ?$$

```
barOMc1 = 1j*10;
mag_L2s_barOMc1 = abs(freqresp(Ls2, barOMc1));
muC_VAL = 1/mag_L2s_barOMc1
```

```
muC_VAL =
78.6313
```

Verify the performance of the controller $C_1(s)$ in every scenario

```
C1s = muC_VAL/s;

Ls1 = C1s*Gs1;
Ls2 = C1s*Gs2;
Ls3 = C1s*Gs3;

[gm, pm, o_gm, o_pm] = margin(Ls1)
```

```
gm =
Inf
pm =
59.9932
o_gm =
Inf
o_pm =
127.1391
```

```
Q1_2Q1_3_FLAG = (pm*o_pm >= 46)
```

```
Q1_2Q1_3_FLAG = logical
   1
```

```

[gm2, pm2, o_gm2, o_pm2] = margin(Ls2)
```

```
gm2 =
Inf
pm2 =
43.3360
o_gm2 =
Inf
o_pm2 =
10
```

```
Q1_2Q1_3_FLAGbis = (pm2*o_pm2 >= 46)
```

```
Q1_2Q1_3_FLAGbis = logical
   1
```

```

[gm3, pm3, o_gm3, o_pm3] = margin(Ls3)
```

```
gm3 =
Inf
pm3 =
57.5820
o_gm3 =
```

```
Inf
o_pm3 =
17.9554
```

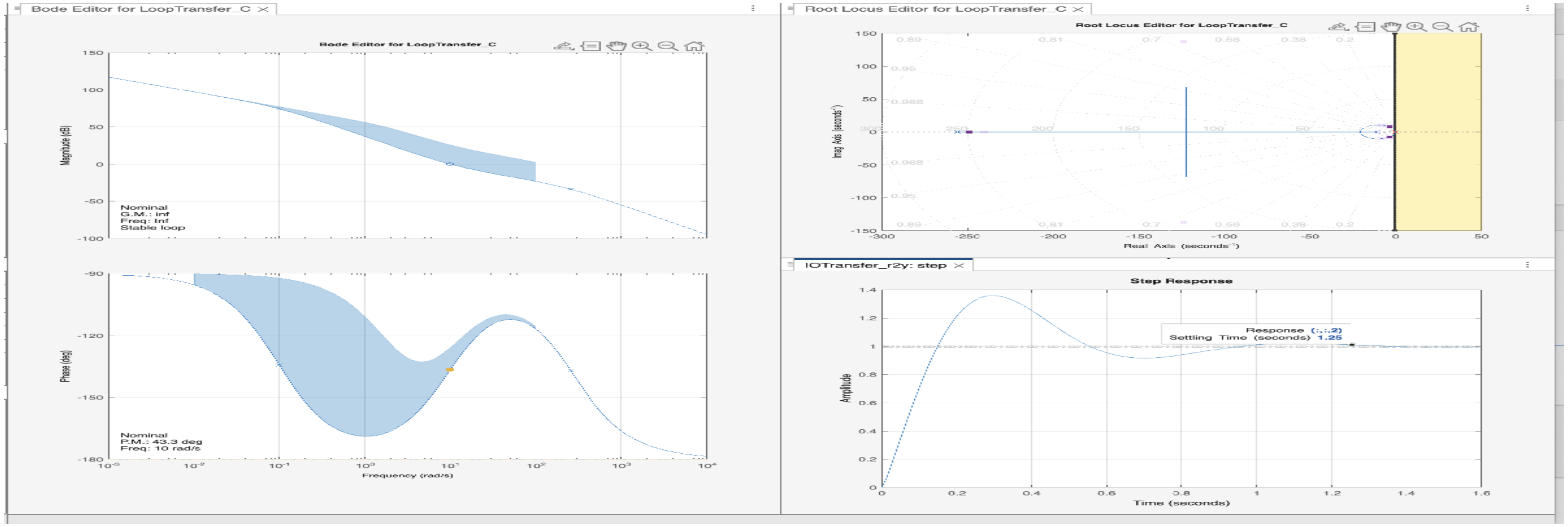
```
Q1_2Q1_3_FLAGter = (pm3*o_pm3 >= 46)
```

```
Q1_2Q1_3_FLAGter = logical
   1
```

In conclusion, the controller $C_1(s)$ allows to fulfil every requirement.

Fine Tuning the Controller using the controlSystemDesigner App

```
% let's create a multi-model structure
Gs = stack(1, Gs1, Gs2, Gs3);

controlSystemDesigner(Gs, C1s)
```



The corresponding controlSystremDesigner session file is **`multimodel_solution_2025_06_06_FA_gr_B_ControlSystemDesignerSession.mat`**